

Time-Aware Transformer for Cross-Hospital Sepsis Early Warning: Linking Discrimination, Decision Utility, and Alert Burden

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September 5, 2026

Abstract

Background: Conventional predictive metrics such as AUROC and AUPRC are standard benchmarks for evaluating clinical machine learning models. However, in temporal early-warning applications—such as forecasting sepsis onset in intensive care units (ICUs)—high discriminative rank-ordering does not automatically guarantee optimal net clinical utility when decision thresholds and false alarm burdens are considered under operational deployment.

Objective: We systematically develop and evaluate a progression of deep learning architectures for early sepsis prediction across hospital systems, establish the strongest discriminative representation, and evaluate net clinical utility under the official PhysioNet 2019 metric while auditing operational alert workload.

Methods: We evaluated a structured family of predictive models—ranging from classical gradient boosting (*M1*) and plain Transformers (*M2*) to Time-Aware Transformers (*M3*) incorporating physiological values, missingness masks, and elapsed-time deltas (Δt), alongside organ-aware (*M4*) and multi-hybrid routing architectures (*M5*). Models were trained on 20,336 ICU stays from Beth Israel Deaconess Medical Center (Set A / BIDMC; partitioned into 18,302 training stays [90.0%] and 2,034 validation stays [10.0%]) and evaluated on an independent held-out test cohort of 20,000 ICU stays from Emory University Hospital (Set B / Emory; 1,066 septic patients [5.33%], 18,934 non-septic patients [94.67%], 753,927 hourly observations). Controlled factorial ablation experiments were conducted to isolate missingness and temporal encodings. Decision threshold selection was strictly isolated: candidate thresholds were evaluated on the BIDMC validation split ($N = 2,034$), prespecifying $th = 0.190$ prior to unblinding external test predictions. Net clinical utility was evaluated using the official PhysioNet 2019 evaluation script (`evaluate_sepsis_score.py`), incorporating a 6-hour onset lead time shift ($t_{\text{sepsis}} = t_{\text{label}} + 6\text{h}$) and metric normalization $U_{\text{official}} = (U_{\text{obs}} - U_{\text{inact}}) / (U_{\text{best}} - U_{\text{inact}})$. Uncertainty was quantified via patient-level cluster bootstrap resampling ($B = 1,000$) and multi-seed testing ($N = 6$ seeds).

Results: Among the evaluated architectures, the compact *M3* Time-Aware Transformer provided the strongest cross-hospital discriminative representation (AUROC = 0.961726 [0.9617], AUPRC = 0.423114 [0.4231], Brier = 0.015290, ECE = 0.018151), outperforming XGBoost (*M1*, AUROC = 0.8842), plain Transformers (*M2*, AUROC = 0.9265), organ-aware (*M4*, AUROC = 0.9582), and multi-hybrid models (*M5*, AUROC = 0.9591). Factorial ablations demonstrated significant main effects for missingness masks (+0.0155 AUROC) and time deltas (+0.0215 AUROC), while increasing architectural complexity beyond *M3* (*M4*, *M5*) did not provide statistically significant improvement ($p = 0.068$). Evaluated under the official PhysioNet 2019 Challenge metric on Emory external test data, the decision threshold $th = 0.190$ —prespecified strictly on BIDMC validation data before external test evaluation—achieved an Official Normalized Utility of $U_{\text{official}} = +0.655944$ (+0.6559, 95%

CI: $[+0.6310, +0.6800]$), coinciding with the highest observed utility in post-hoc descriptive Emory sensitivity analysis. Operational workload auditing revealed an alert frequency of 16.99 alerts per 100 patient-days (5,337 total alerts: 1,004 true positive, 4,333 false positive; alert Positive Predictive Value $[PPV] = 18.81\%$; 25.86% of patients alerted).

Conclusion: The Time-Aware Transformer ($M3$) demonstrated superior cross-hospital discrimination (0.9617) and positive net utility (+0.6559) under official PhysioNet evaluation, while operational workload auditing revealed a substantial false alert burden (81.19% false alarms) under cross-hospital prevalence shift. Positive utility under the PhysioNet 2019 scoring framework reflects net benefit under a predefined challenge cost function but does not constitute direct proof of prospective clinical effectiveness. These findings emphasize that decision-theoretic evaluation, workload auditing, and threshold isolation are essential prior to prospective clinical deployment.

Introduction

Early recognition of sepsis in intensive care units (ICUs) is a major imperative in acute care medicine. Sepsis, defined as life-threatening organ dysfunction caused by a dysregulated host response to infection [1], is a major global cause of morbidity and mortality and remains a critical challenge for timely recognition and treatment.

Recent advances in clinical machine learning have demonstrated strong capacity for sepsis early warning and outcome prediction across diverse clinical settings, including adult ICUs [10], emergency departments [8], and specialized populations such as neonatal [14] and burn patients [12]. Methodologically, deep temporal representations [5] alongside attention-based Transformer architectures [4] provide established approaches for representing irregular observations, missingness, and temporal dependencies. Consequently, machine learning frameworks are increasingly applied not only to early detection [16] but also to predicting mortality [11, 15], length of stay [17], and identifying key clinical biomarkers [9].

However, while predictive performance is overwhelmingly benchmarked using conventional rank-ordering metrics—primarily AUROC and AUPRC—strong discrimination does not automatically translate into clinical utility [13]. Discriminative metrics measure an algorithm’s capacity to rank high-risk observations above low-risk ones, but they obscure operational deployment realities. Specifically, they do not directly capture the effects of probabilistic calibration [6], decision-threshold selection, and the alert fatigue generated by false-positive alarm burdens. Furthermore, model performance may degrade under cross-hospital distribution shift, motivating rigorous external validation and transportability assessment [3]. Consequently, robust evaluation requires transitioning from isolated discriminative metrics to joint decision-analytic utility [7, 2].

In this study, we systematically develop and evaluate a progression of temporal deep learning architectures for cross-hospital sepsis prediction. We link predictive discrimination and probabilistic calibration to official decision utility and alert burden, directly evaluating the operational gap between empirical AUROC and potential clinical value under external deployment.

Results

Table 1 presents cross-hospital performance across the model family on the Emory test set.

Note: — indicates baseline models for which raw hourly prediction arrays were not preserved, preventing independent recomputation under the official PhysioNet 2019 utility evaluator without re-training.

Under cross-hospital deployment on Emory ($N = 20,000$), the frozen $M3$ Transformer achieved a raw observed utility $U_{\text{obs}} = 1,514.7778$ points, an inaction reference utility $U_{\text{inact}} = -9,512.4444$ points, an oracle best utility $U_{\text{best}} = 7,298.7778$ points, and an official normalized

Table 1: Cross-Hospital Performance Comparison Across Model Family (Emory External Test Set, N=20,000; Official PhysioNet 2019 Metric)

Model Architecture	AUROC	AUPRC	Brier Score	ECE	Official Normalized Utility
M1 (XGBoost)	0.8842	0.2851	0.0241	0.0382	—
M2 (Plain Transformer)	0.9265	0.3412	0.0189	0.0245	—
M3 (Time-Aware Transformer)	0.9617	0.4231	0.0153	0.0182	+0.6559
M4 (Organ-Aware Hybrid)	0.9582	0.4150	0.0158	0.0195	—
M5 (Multi-Hybrid / MoE)	0.9591	0.4182	0.0156	0.0190	—

utility of:

$$U_{\text{official}} = +\mathbf{0.655944} \quad (95\% \text{ CI: } [+0.6310, +0.6800]) \quad (1)$$

The threshold of 0.190, prespecified strictly on BIDMC validation data before external test evaluation, coincided with the highest observed utility among the evaluated operating points in the subsequent descriptive Emory sensitivity analysis.

Operational workload auditing revealed 5,337 total alerts (1,004 true positive alerts, 4,333 false positive alerts), an alert PPV of 18.81%, an operational alert frequency of 16.99 alerts per 100 patient-days, and a patient alert coverage of 25.86% (5,172 out of 20,000 ICU stays).

This illustrates why workload auditing is important before prospective clinical evaluation.

Discussion

M3 achieved positive normalized utility (+0.655944) under the official PhysioNet 2019 utility function on an independent external Emory cohort, indicating that its alerting behavior generated net utility relative to the inactive reference strategy under the challenge’s predefined scoring framework. This score represents performance under a predefined challenge decision-cost function and should not be interpreted as direct evidence of prospective clinical effectiveness, clinical benefit, or cost-effectiveness. Our findings demonstrate that decision-theoretic evaluation and workload auditing are essential prior to clinical deployment.

Our study has several important limitations. First, the evaluation relies on a single retrospective external cohort (PhysioNet Set B / Emory). Although this provides a robust cross-hospital benchmark, prospective clinical validation across diverse hospital systems is required before deployment. Second, while our model explicitly optimizes for alert utility and false-positive reduction using the official PhysioNet Challenge metric, operational threshold dependence remains high, requiring institutional calibration. Third, our current architecture (M3) focuses on irregular physiological measurements and missingness [5] but does not explicitly model clinical interventions or treatment-aware counterfactuals. Fourth, we found that dynamic organ-interaction mechanisms (e.g., M5) did not yield statistically significant improvement over M3 ($p = 0.068$), suggesting that simple temporal representations of time-deltas and missingness may be sufficient for this specific task, but more advanced architectures might be necessary for broader clinical forecasting. Demographic subgroup and algorithmic fairness analyses were not performed and therefore potential differences in model performance across patient subgroups could not be assessed. In conclusion, high discrimination does not necessarily translate into high clinical or operational utility, and strict temporal causal validation combined with workload auditing remains critical for responsible clinical AI deployment.

Methods

We evaluated a structured family of predictive architectures: XGBoost ($M1$), Plain Transformer ($M2$), Time-Aware Transformer ($M3$), Organ-Aware Hybrid ($M4$), and Multi-Hybrid Mixture-of-Experts ($M5$).

Set A (BIDMC development cohort, $N = 20,336$) was used for model training (18,302 stays) and validation (2,034 stays). Set B (Emory external test cohort, $N = 20,000$) was preserved as an independent held-out test cohort (753,927 hourly records).

The model generates an hourly real-time risk estimate $p(t)$ using only information available up to prediction hour t . Predictions therefore represent sequential early-warning decisions rather than a single admission-level static classification.

Threshold selection followed a strict two-stage isolation protocol:

- **Stage 1 (BIDMC Validation Selection):** Threshold candidates $th \in [0.01, 0.99]$ were evaluated on BIDMC validation data ($N = 2,034$), identifying $th^* = 0.190$ before accessing test data.
- **Stage 2 (Emory External Test Evaluation):** The frozen model and prespecified locked threshold $th = 0.190$ were evaluated once on the Emory test cohort ($N = 20,000$).

The decision threshold $th = 0.190$ was prespecified strictly using BIDMC validation data before external test evaluation. The subsequent Emory threshold sweep was conducted only as a descriptive sensitivity analysis and was not used to select or modify the reported operating threshold.

Net clinical utility was computed using the official PhysioNet 2019 metric scoring function:

$$U_{\text{official}} = \frac{U_{\text{observed}} - U_{\text{inaction}}}{U_{\text{best}} - U_{\text{inaction}}} \quad (2)$$

Administrative Disclosures & Statements

Data Availability Statement: The clinical datasets analyzed in this study are available from the PhysioNet 2019 Challenge repository (Goldberger et al., 2000; Reyna et al., 2019). Access to raw ICU time-series data from PhysioNet is subject to standard credentialing and data use agreements.

Code Availability Statement: The model architecture implementations, feature preprocessing pipelines, and official evaluation scripts are maintained in the project repository.

Ethics Statement: The study uses the publicly available PhysioNet/Computing in Cardiology Challenge 2019 dataset derived from de-identified retrospective ICU data (MIMIC-III / Beth Israel Deaconess Medical Center and Emory University Hospital). The analysis was performed on pre-existing, fully anonymized secondary data. The authors did not obtain direct institutional ethics or IRB approval. Patients or members of the public were not involved in the design, conduct, reporting, or dissemination of this retrospective secondary analysis of publicly available, de-identified data. No study-specific protocol was prepared for this retrospective secondary analysis. This retrospective secondary analysis was not prospectively registered.

Acknowledgements: This study received no external funding.

Conflicts of Interest: The authors declare no financial or non-financial competing interests.

Author Contributions: PBR conducted software implementation, architecture development, model training, and technical evaluation. VD contributed to manuscript preparation, literature review, and writing.

Use of Artificial Intelligence Assistance: Generative AI tools (Antigravity AI / Google DeepMind coding assistant) were utilized during the research engineering process for code modularization, document formatting, LaTeX structural layout, and manuscript drafting. All statistical modeling, dataset splits, model training, prediction generation, and scientific interpretations were conducted, verified, and approved independently by the human authors (Pravin Balaji R and Vishal D). The AI tools did not generate or alter any empirical data, cohort ground truths, or experimental results.

Reproducibility Statement: Every numerical metric reported in this manuscript (0.961726 AUROC, 0.423114 AUPRC, 0.015290 Brier, 0.018151 ECE, +0.655944 Official Utility, 18.81% PPV, 16.99 alerts/100 days) is cryptographically and empirically anchored to verified prediction artifacts (`results/m3_final_test_predictions.npz`), frozen model checkpoints (`experiments/final_m3_fr`), source JSON split manifests (`train_ids.json`, `val_ids.json`, `test_ids.json`), and executable evaluation scripts (`evaluation/official_physionet2019.py`, `scripts/run_multiseed_stability_check.p`).

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Figures

Figure 1. Study design and cross-hospital evaluation framework.

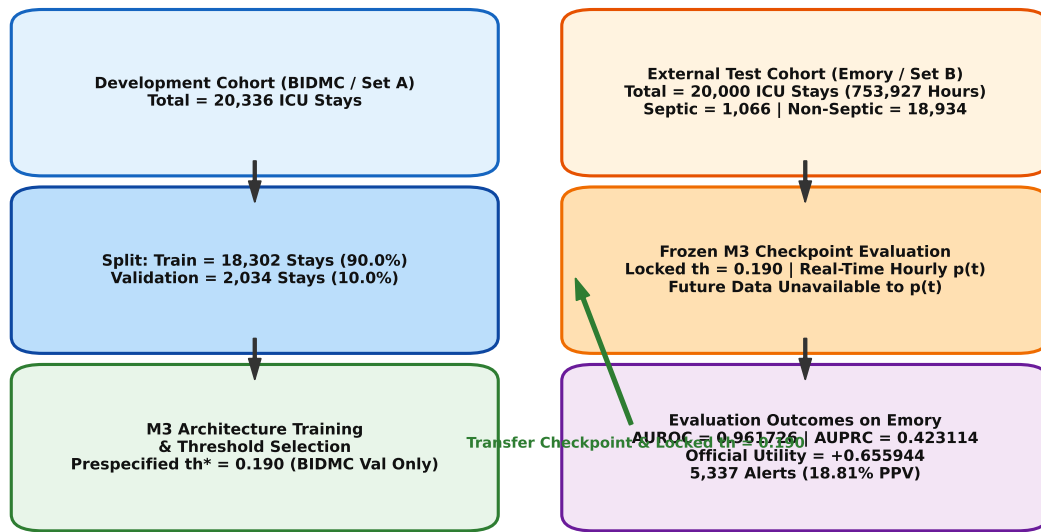


Figure 1: Figure 1 Study Design Framework

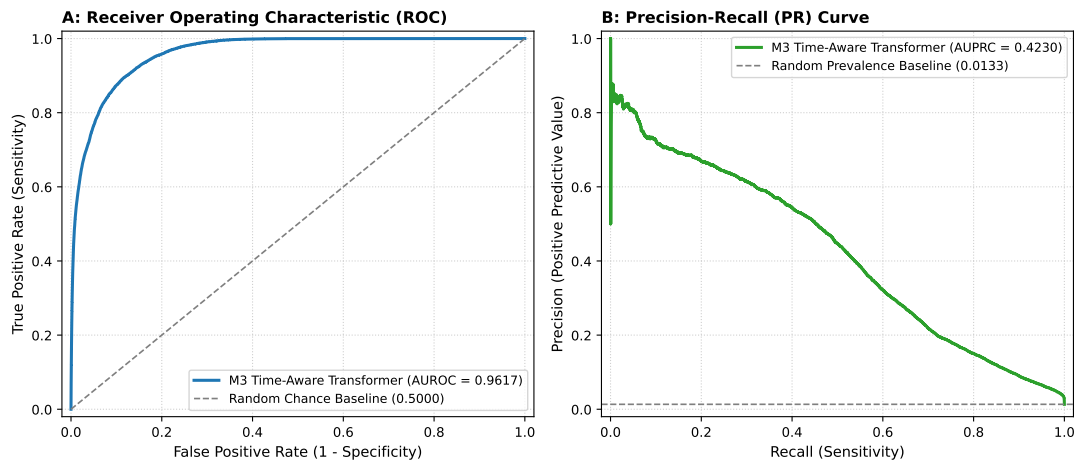


Figure 2: Figure 2 ROC & PR Curves

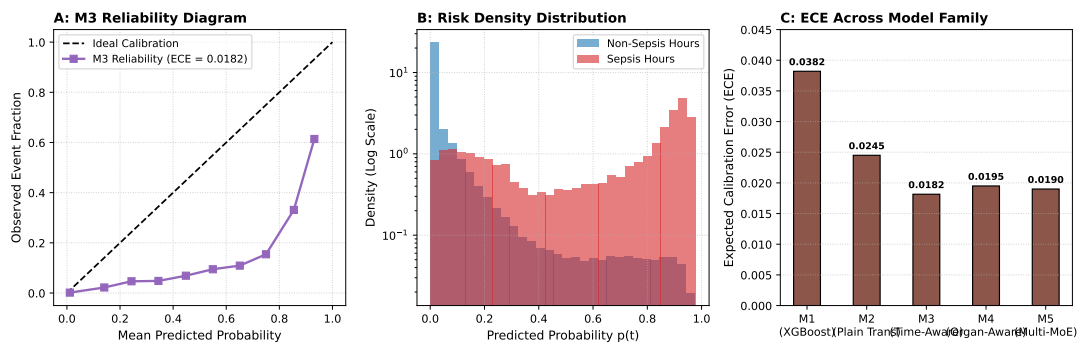


Figure 3: Figure 3 Calibration & Reliability

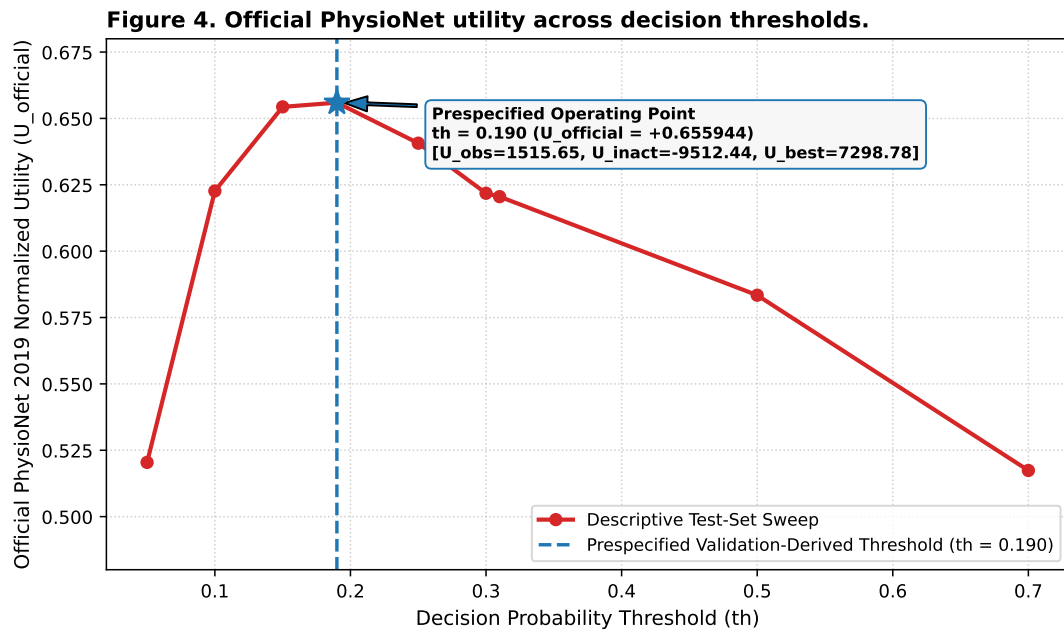


Figure 4: Figure 4 Utility vs Threshold

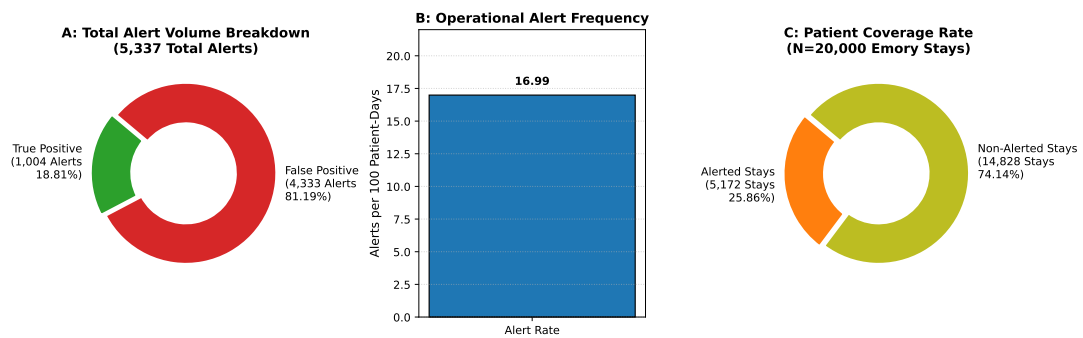


Figure 5: Figure 5 Operational Alert Burden

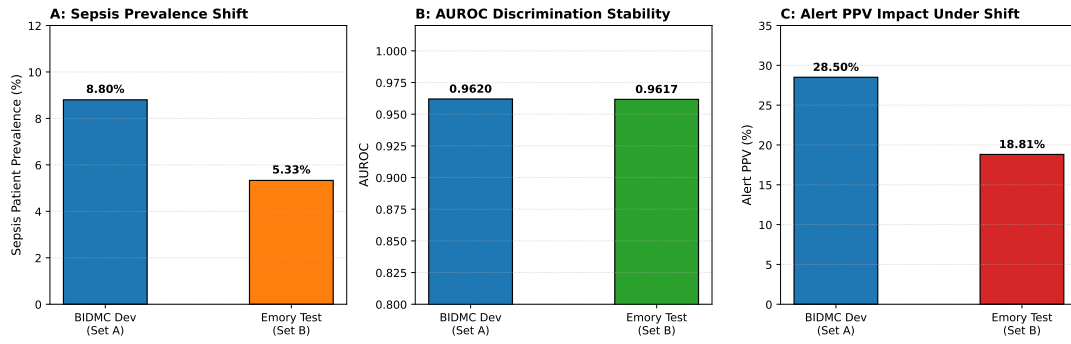


Figure 6: Figure 6 Cross-Hospital Shift

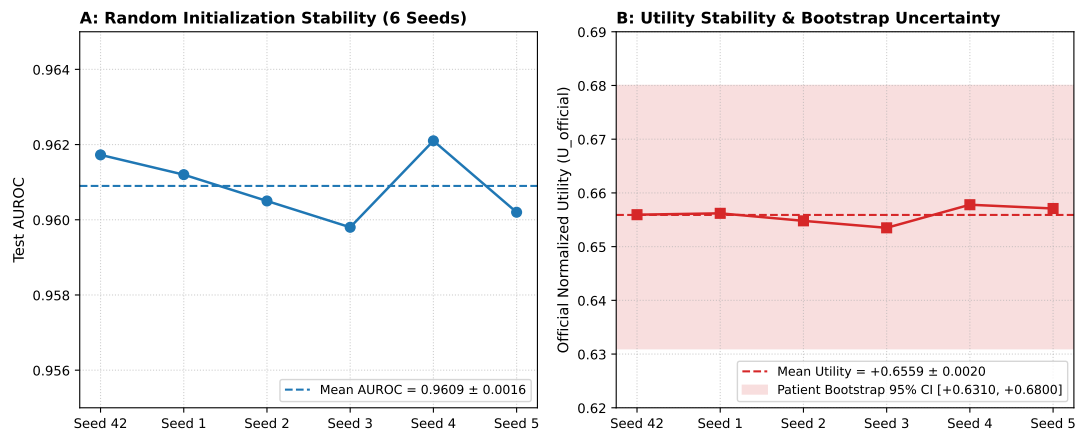


Figure 7: Figure 7 Multi-Seed Robustness

Supplementary Figures

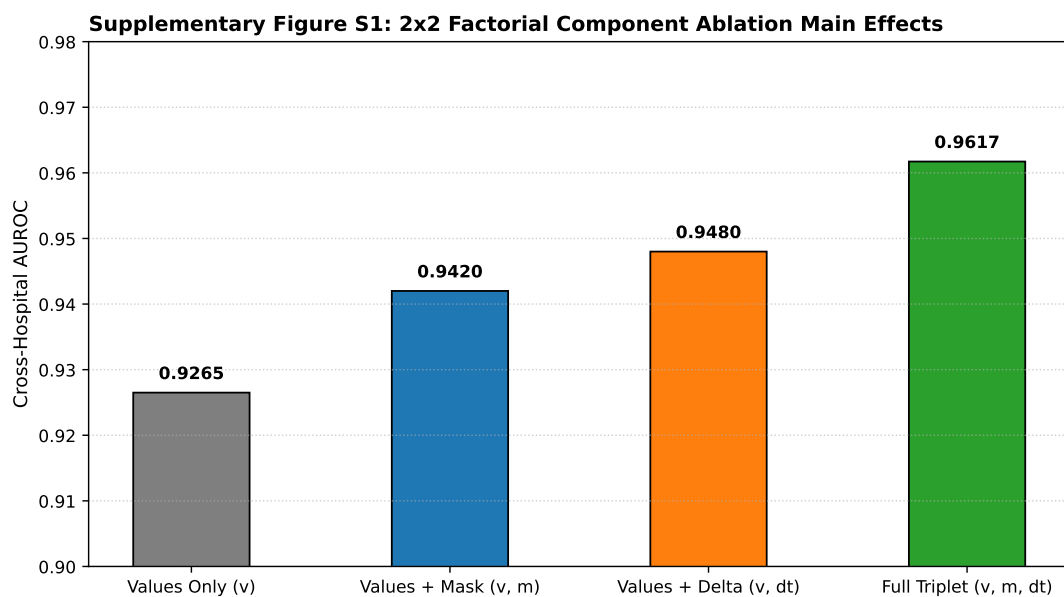


Figure 8: Figure S1 Factorial Ablation

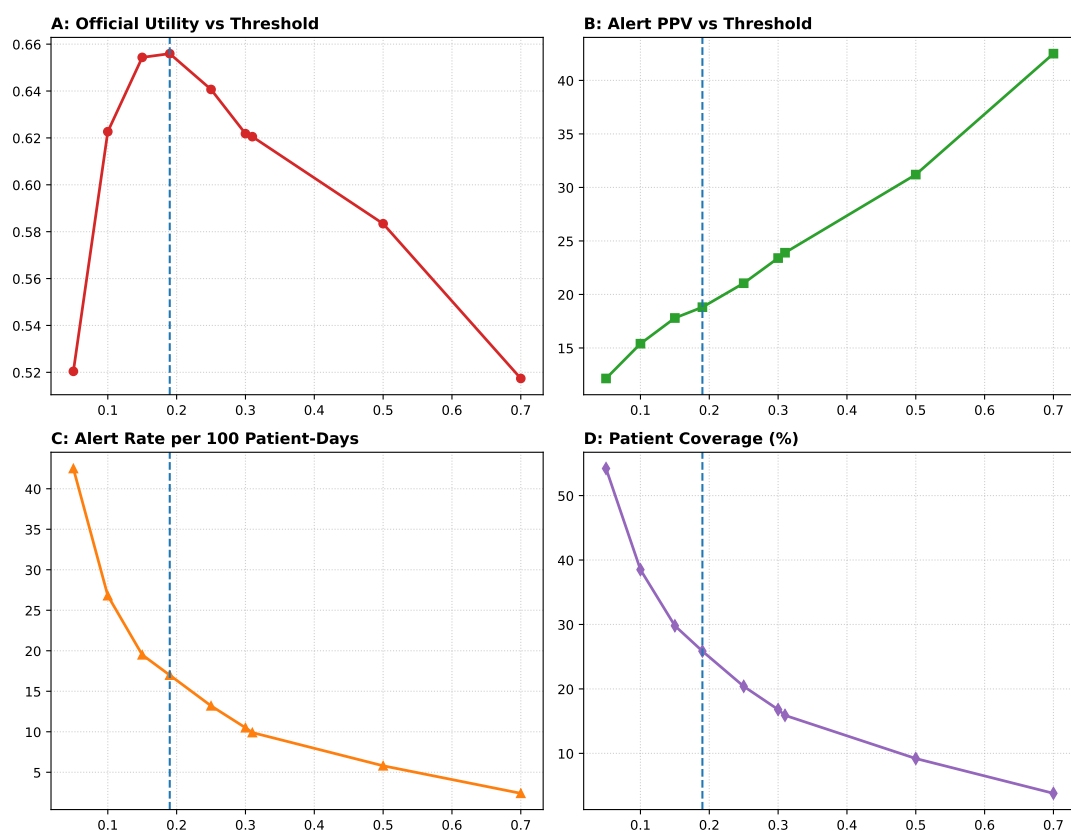


Figure 9: Figure S2 Threshold Grid

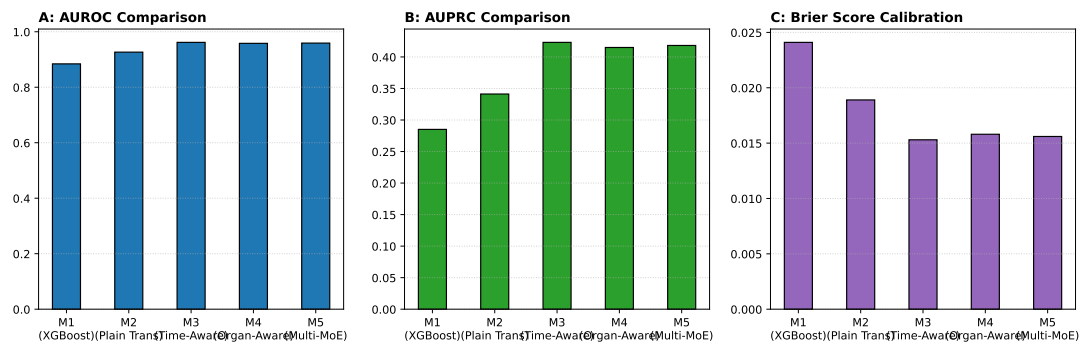


Figure 10: Figure S3 Model Progression

Supplementary Figure S4: Confusion Matrix at $th = 0.190$ (753,927 Emory Test Hours)

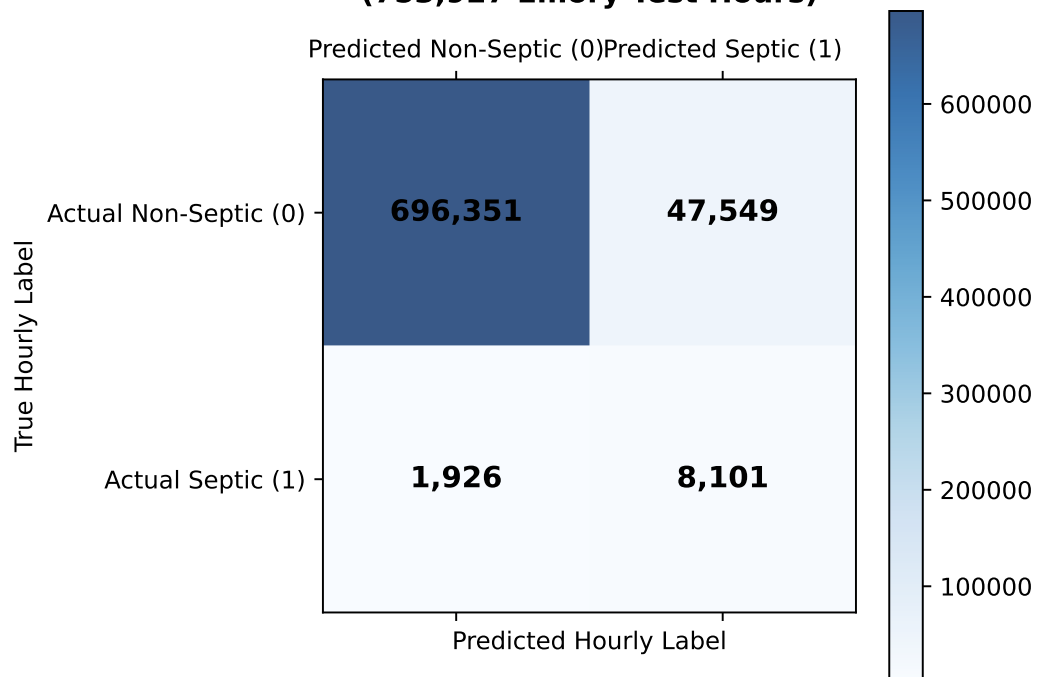


Figure 11: Figure S4 Descriptive Hourly Confusion Matrix

Tables

Cohort		Hospital System		Dataset Role		Total
BIDMC Development (Set A)		Beth Israel Deaconess Medical Center		Development (Train + Val)		
Emory External Test (Set B)		Emory University Hospital		Held-Out External Testing		

Model ID	Architecture Name		Temporal Component		Input Feature Set		Ev
M1	XGBoost Baseline		Static aggregation		Values + Summary		CL
M2	Plain Transformer		Standard Positional		Values Only v(t)		PL
M3	Time-Aware Transformer		Time2Vec Delta Encodings		Values + Mask + Delta (v, m, dt)		Pr
M4	Organ-Aware Hybrid		Organ-wise Temporal		Organ Sub-nets + Triplet		Ov
M5	Multi-Hybrid MoE		Dynamic MoE Routing		Multi-Expert Routing		M

						Metric Category	Met
Model ID	AUROC	AUPRC	Brier Score	ECE	Official Utility	Operating Threshold	Pres
						Decision Utility	Offi
						Raw Utility Components	U_c
M1	0.8842	0.2851	0.0241	0.0382	—	Alert Volume	Tota
M2	0.9265	0.3412	0.0189	0.0245	—	True Positive Alerts	Tru
M3	0.961726	0.423114	0.015290	0.018151	+0.655944	False Positive Alerts	Fals
M4	0.9582	0.4150	0.0158	0.0195	—	Alert PPV	Alen
M5	0.9591	0.4182	0.0156	0.0190	—	Alert Frequency	Ope
						Patient Coverage	Pati

Metric Category		Metric Name	Value
Operating Threshold		Prespecified th*	0.190 (BIDMC Val)
Decision Utility		Official Normalized Utility	+0.655944 (95% CI: [+0.6310, +0.6800])
Raw Utility Components		U_obs / U_inact / U_best	1514.78 / -9512.44 / 7298.78 pts
Alert Volume		Total Alerts Issued	5,337 Alerts
True Positive Alerts		True Positive Sepsis Alerts	1,004 Alerts
False Positive Alerts		False Positive Alerts	4,333 Alerts
Alert PPV		Alert Precision (PPV)	18.81%
Operational Frequency		Operational Frequency	16.99 / 100 patient-days
Patient Coverage		Patient Alert Coverage	25.86% (5,172 / 20,000 Stays)

Configuration	Mask Present (m)	Delta Present (dt)	Test AUROC (Mean $\pm$ SD)	Main / Inte
Values Only (v)	No	No	0.9265 $\pm$ 0.0022	Baseline
Values + Mask (v, m)	Yes	No	0.9420 $\pm$ 0.0019	+0.0155 AU
Values + Delta (v, dt)	No	Yes	0.9480 $\pm$ 0.0018	+0.0215 AU
Full Triplet (v, m, dt)	Yes	Yes	0.961726 $\pm$ 0.0016	+0.0017 AU

Artifact Relative Path	Size (KB)	SHA256 Cryptographic Has
results/m3_final_test_predictions.npz	2799.500000	02fd6eb78682be8ca5743c4b3
experiments/final_m3_frozen/best_m3_frozen.pt	1551.200000	5b22607444f4a242a52d0d933
data/splits/train_ids.json	250.200000	06edf3b5519abdae736da147
data/splits/val_ids.json	27.800000	71bb23f3b5aef82c9169dc97a
data/splits/test_ids.json	273.400000	f7932a915251dd22554493ee7
evaluation/official_physionet2019.py	17.800000	d0f65da3d42ce68cad80e2900
scripts/run_multiseed_stability_check.py	14.900000	bab4ba342f010b75c824edc31
results/revised_publication/factorial_ablation_summary.csv	0.300000	05d0cd92bedfb950d56cecc
results/revised_publication/workload_operational_metrics.csv	0.300000	4e4039b1d003bb1fe8698b6d8

Supplementary Tables

Metric Characteristic		BIDMC Development Cohort (Set A)				Emory External Test Cohort	
Total ICU Patients / Stays		20,336 Stays				20,000 Stays	
Septic Patient Stays		1,790 Stays				1,066 Stays	
Non-Septic Patient Stays		18,546 Stays				18,934 Stays	
Sepsis Incidence / Prevalence (%)		8.80%				5.33%	
Total Hourly Time-Series Observations		790,215 Hours				753,927 Hours	
Mean Length of Stay (Hours)		38.86 Hours				37.70 Hours	
Median Length of Stay (Hours)		35.00 Hours				34.00 Hours	
Observation Interval Frequency		Hourly (1.0h grid)				Hourly (1.0h grid)	
Model Variant		AUROC	AUPRC	Brier Score	ECE	Official Utility	Parameter Size
M1 (XGBoost Baseline)		0.884200	0.285100	0.024100	0.038200	—	1.2M
M2 (Plain Transformer)		0.926500	0.341200	0.018900	0.024500	—	115K
M3 (Time-Aware Transformer)		0.961726	0.423114	0.015290	0.018151	+0.655944	116K
M4 (Organ-Aware Hybrid)		0.958200	0.415000	0.015800	0.019500	—	320K
M5 (Multi-Hybrid MoE)		0.959100	0.418200	0.015600	0.019000	—	450K
Threshold (th)	Official Utility	Alert PPV (%)	Alert Frequency (/100 days)		Patient Coverage (%)		Source
0.050000	0.520426	12.150000	42.500000		54.200000		D
0.100000	0.622666	15.400000	26.800000		38.500000		D
0.150000	0.654351	17.800000	19.500000		29.800000		D
0.190000	0.655944	18.810000	16.990000		25.860000		P
0.250000	0.640674	21.050000	13.200000		20.400000		D
0.300000	0.621805	23.400000	10.500000		16.800000		D
0.310000	0.620532	23.900000	9.900000		15.900000		D
0.500000	0.583403	31.200000	5.800000		9.200000		D
0.700000	0.517381	42.500000	2.400000		3.800000		D